

The Kabuki Papers

Kabuki Syndrome science, illuminated by AI

The Kabuki Papers: an overview

Rare diseases are the ultimate "news desert" - research is happening, but very little of it is translated into a form accessible to the families struggling to understand what's going on with their child.

Since my son was diagnosed with a rare disease, Kabuki Syndrome, I have spent hours attempting to push my way through dense academic papers that described a promising experimental therapy, or offered insights into this complex disorder. I never really made to the end of a single one!

AI changed that for me. Just pasting a link into ChatGPT was enough to shine a light into the shadows that had been befuddling me and thousands of others. Soon my understanding of Kabuki was expanding rapidly.

During a recent career break, I started thinking about coming up with a project to start learning more deeply about AI, and it seemed to make the most sense to put Kabuki at the center of that. After a few iterations, I came on the idea of

creating a newsletter (<https://www.thekabukipapers.org>) about the science behind Kabuki that would be entirely AI-generated. I would act as editor, not really knowing at first what that would mean.

In the end it entailed a mix of duties that combined my programming side with my writing side (I'm a software developer who's committed occasional acts of journalism). First was the decision to have each post be a direct translation of a single paper. Choosing which paper to translate would be a weekly, essentially manual undertaking. A second one, harder to easily define, would be to find my readers, and continuously seek to understand and serve their interests and concerns.

Copying and pasting a link into ChatGPT provides a splendid summary, but I saw that any AI summary has two essential flaws. First, hallucinations. There is always the doubt there that what you're reading can go off the rails at any minute and produce falsehoods. Secondly, completeness. Something is always being left out in any summary - but it's often unclear what it is. I saw the possibility of mitigating these two concerns as an opportunity for my offering to add value.

In the end, both are neatly addressed by creating an automated pipeline that includes an initial step where the original paper is reduced into a series of several dozen claims. These serve as the source of truth for the eventual translation. This allowed me to do two things.

First, to be able to tell the AI in a subsequent step to include all claims in the eventual translations, increasing the probability of completeness.

Second, to provide a feature where the reader can click on individual sentences to see the passages in the original paper they were based on, increasing

confidence in accuracy (see this in action here: <https://monosnap.ai/file/1FwVwDDDDyBDS4LUvl81HZsCMyvPGdb>).

A paramount "reader concern" that I needed to address was, of course, readability. Reaching this goal a large part of my time in this project. I spent several weeks experimenting with prompting approaches and several LLM providers, seeking to find a balance between scientific thoroughness and what I came to think of as "sparkle" - that quality in writing that keeps you reading until the end.

The eventual pipeline had a few steps:

1. extract structure from the paper. There was some nuance here, but this mostly came down to a list of core paragraphs, excluding some of little interest to the reader, like those related to technical methods.
2. translate the core paragraphs into a json file listing all "claims" made by the

```
{
  "claim": "BDNF, CREB, SDF-1 and CXCR4 are important in neurogenesis in adult animals.",
  "basedOnText": [
    "Certain neurotrophic factors and neurotransmitter receptors, including brain-derived neurotrophic factor (BDNF), cyclic adenosine monophosphate response element-binding protein (CREB), stromal cell-derived factor-1 (SDF-1) and its specific receptor, C-X-C motif chemokine receptor 4 (CXCR4), are important in neurogenesis in adult animals."
  ],
  "reference_id": "1"
},
```

paper. Here's what what one claim looked like, as an example:

3. use those claims to create the editorial components of the post: the headline, subheadline, actual text, illustrations, pull quotes, tags, category, term definitions etc. This was the most involved challenge overall.

At any step, any of the following LLMs could be used: ChatGPT, Claude, Gemini, Kimi, MiniMax, DeepSeek or a local Gemma instance. I built a flexible

invocation framework to allow easy experimentation, including the ability to run prompts on all of them at once. This allowed me to hone in more efficiently on which one was capable of creating the voice I wanted. This process was definitely a next-gen "editorial" function, combining empathy for my readers, wrangling API endpoints, gaining understanding of each LLM's "feel" and writing efficient, agile pipeline code.

Maintaining the ability to tweak the voice was paramount. This was especially important when I shared my first post.

The user I had in mind during development was like myself: someone who had steeped themselves in the science to a certain degree. As a result, there was a level of scientific terminology I let stay in the translations. Some, but not too much. Enough, I thought, to provide avenues for further investigation to the reader without bogging down their reading experience.

When I shared the first post, written with that tone, in the Kabuki Facebook group, I immediately recognized that I needed to tone it down. My readers, I saw, were mostly overwhelmed with caring for their sick children and weren't quite as deep in the weeds as I had gotten. I added language to my core prompt to use fewer (and fewer) technical terms. This made for a much cleaner read overall.

In the end I believe the project has been successful - feedback is positive in the group and several dozen have signed up and read each post.

What's next? In a sense the project has successfully found a growing audience and is "done" - but in another sense it's just the beginning. Having a database of claims across papers allows a galaxy of possibilities - from creating new ways of explaining (quizzes, creative illustrations, slideshows) to combining multiple papers to find trends and broader insights.

